Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| Title of Proposed Technology | Microgravity-Integrated Handheld 3D Bioprinting Platform for Regenerative Medicine Applications | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1)] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | Bio · Medical | | Mechanical · Materials | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Bio-processes / Bio-Equipment | Bio-engineering Technologies | Precision Manufacturing Machinery | Injection Molding Machines |
| Concept of Proposed Technology | The proposed technology is a **microgravity-integrated 3D bioprinting platform** designed to fabricate biologically functional tissues under simulated spaceflight conditions and enable **in situ regenerative treatment during space missions**.<br><br>As human space activities transition toward long-duration habitation in Low Earth Orbit (LEO) and beyond, physiological degradation—including muscle atrophy, cardiovascular dysfunction, and impaired tissue regeneration—has emerged as a critical challenge.[1,2] While ground-based 3D tissue models have contributed to understanding these phenomena, conventional terrestrial fabrication is fundamentally limited by gravity-driven deformation, scaffold dependency, and altered fluid dynamics that diverge from actual space environments.[3]<br><br>This project introduces the concept of **"Aerospace 3D Bioprinting,"** in which microgravity is treated not as a constraint but as an enabling manufacturing condition.[4] By integrating a 3D bioprinter with a **6-degree-of-freedom (6-DOF) microgravity simulation system**, the platform | | | |

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| | enables tissue fabrication in an environment that more closely mimics in-space biological conditions.[4]<br><br>A defining feature of the proposed technology is its **dual-mode architecture**:<br><br>• A **stationary microgravity-integrated bioprinting system** for controlled tissue fabrication and evaluation<br>• An **ultra-lightweight, detachable handheld bioprinter module** that enables direct in situ bioprinting onto damaged tissues, allowing rapid response to injuries in confined or emergency situations during space missions[5]<br><br>The platform utilizes **ECM-based bioinks optimized for high-speed crosslinking** through temperature- and UV-based dual curing. This material strategy ensures shape retention and structural stability under microgravity while preserving high cell viability and biological functionality.<br><br>Through comparative analysis of tissues fabricated under simulated microgravity and terrestrial gravity, the project aims to elucidate the biological and biochemical effects of space manufacturing conditions and establish a foundation for future in-space regenerative medicine.[6] |
| **Rationale for Joint R&D Project** | Despite advancements in national aerospace capabilities, there remains a **structural gap in experimental infrastructure and validated biological data** related to tissue behavior and regeneration under microgravity conditions. Addressing this gap requires a joint R&D framework that combines **space-environment experimentation expertise** with **advanced bioprinting system development capabilities**.<br><br>From the research institute's perspective, international joint R&D is necessary to:<br><br>• Expand experimentally validated datasets on cellular behavior, tissue maturation, and functional recovery under microgravity<br>• Establish reproducible evaluation frameworks linking space environmental factors with biological outcomes<br>• Create a translational pathway from ground-based microgravity simulation to future in-orbit experimentation<br><br>Collaboration with Korean SMEs enables translation of accumulated experimental knowledge into **engineering-realistic, deployable bioprinting systems**, while simultaneously reducing development risk and accelerating readiness for future space demonstration. This joint approach supports not only technology development but also the creation of a validated infrastructure capable of verification in real space environments. |
| **Key Words** | 1. 3D bioprinter<br>2. handheld bioprinting<br>3. microgravity<br>4. Myogenesis |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level ( 3) | End of the Project | Level ( 5 ) |
|---|---|---|---|---|

**R&D Objectives**

**Overall Objective**

To develop a **compact, lightweight, and modular microgravity-integrated 3D bioprinting system** with handheld capability, validate its performance through biological tissue fabrication and evaluation, and advance the technology from **TRL 3 to TRL 5**.

**Specific Objectives**

- Develop a compact 3D bioprinter compatible with a 6-DOF microgravity simulation system
- Design and validate a detachable handheld bioprinting module for in situ tissue application
- Develop ECM-based bioinks with high-speed crosslinking suitable for microgravity environments
- Fabricate and evaluate muscle tissue constructs under simulated microgravity[1,2]
- Quantitatively assess tissue viability, morphology, and functional indicators relative to ground controls

**R&D Target Specifications**

**System-Level Targets**

- Handheld bioprinter module weight: **≤ 1 kg**
- Total integrated system weight: **≤ 4 kg**
- Modular, detachable, and compact system architecture suitable for confined environments
- Construction of **one 3D bioprinting system unit** compatible with a 6-DOF microgravity simulator

**Bioink and Printing Performance**

- Crosslinking time: **≤ 1 minute**
- Post-printing cell viability: **≥ 90% (+)** [3]
- Shape fidelity maintained under continuous 6-DOF rotation

**Tissue Fabrication and Evaluation**

- Fabrication of muscle tissue constructs with representative dimensions
  - Primary target: **8 × 4 × 4 mm**
  - Additional evaluation sizes within scalable millimeter range for structural stability testing
- Quantitative evaluation including:
  - Cell viability and proliferation
  - Histological integrity
  - Structural stability during culture
  - Functional responsiveness compared to terrestrial controls

**R&D Plan**

**Research Approach**

- **Platform Development**:
  Development of a 3D bioprinting system mountable on a 6-DOF microgravity simulator (RPM/Clinostat) to enable tissue fabrication under simulated microgravity conditions
- **Mobility and Emergency Use**:
  Development of a detachable handheld bioprinter enabling direct, in situ application to damaged tissues during emergency scenarios

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**Key Tasks and Methodologies**

*Mechanical Engineering*

- Lightweight frame and drive system design optimized for microgravity simulation
- Modular, foldable, and detachable system architecture
- Stable printing performance under continuous 6-DOF rotation

*Material Science*

- Development of tissue-specific ECM-based bioinks
- Optimization of GelMA/HAMA-based formulations
- Dual crosslinking (temperature and UV/VIS) for rapid curing and shape retention

*Biological Validation*

- Assessment of post-printing cell viability and proliferation
- Histological analysis of tissue maturation
- Comparative functional evaluation versus terrestrial gravity–fabricated tissues

**Expected Milestones & Timeline**

**Year 1 (Phase I)**

- Design and fabrication of handheld bioprinter and detachable module
- Identification and preliminary evaluation of high-speed crosslinkable ECM bioinks

**Year 2 (Phase II)**

- Integration of 3D bioprinting system with 6-DOF microgravity simulator
- Optimization of bioprinting parameters for microgravity conditions

**Year 3 (Phase III)**

- Finalization of integrated prototype
- Comprehensive performance verification
- Quantitative evaluation of tissue fabricability, viability, histological maturity, and functional response under simulated microgravity

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | The research institute will lead **biological validation and microgravity-specific evaluation**, including:<br><br>• Development and optimization of ECM-based bioinks for microgravity printing<br>• Execution of bioprinting experiments under 6-DOF simulated microgravity<br>• Biological evaluation of fabricated tissues, including viability, histology, and functional response<br>• Comparative analysis between microgravity-fabricated and terrestrial control tissues<br>• Establishment of experimental protocols and evaluation standards for space bioprinting |
| **Expected R&R of Korean SME Partner** | The Korean SME partner will lead **engineering development and system integration**, including:<br><br>• Development of the lightweight handheld bioprinting module<br>• Engineering of the integrated 3D bioprinting system compatible with 6-DOF simulation<br>• Design of modular, detachable, and compact system architecture<br>• System assembly, testing, and performance optimization<br>• Collaboration on prototype validation and preparation for future space demonstration |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | **Technological Outcomes**<br><br>• Development of a **dual-mode microgravity bioprinting platform** combining stationary and handheld operation<br>• Establishment of high-speed crosslinkable ECM bioink protocols for space environments<br>• Generation of validated data on tissue fabrication and maturation under microgravity<br><br>**Commercial and Strategic Impact**<br><br>• Entry into the emerging **"Space-Bio" convergence industry**<br>• Foundation for commercialization in space habitation, orbital stations, and deep-space missions<br>• Dual-use terrestrial applications, including emergency medicine, disaster response, and point-of-care regenerative treatment<br><br>**Industrial and Societal Impact**<br><br>• Strengthening of national technological sovereignty in space medicine and bio-manufacturing<br>• Creation of a ground-based microgravity bioprinting testbed supporting global collaboration<br>• Long-term contribution to sustainable human space exploration and remote regenerative medicine |

## 6. References


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